Dr. Navneet Munoth

Associate Professor, Department of Architecture & Planning, MANIT Bhopal

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Architect and urban planner (PhD, IIT Roorkee) specialising in urban renewal and redevelopment, urban conservation, physical planning, remote sensing and GIS, and the Internet of Things in the built environment. Research output: 18 journal papers, 22 conference papers, 9 book chapters, and 3 books.

Research projects, consultancy & patent

Ongoing2023–2025

Design principles and indigenous construction techniques of the Mahakaal Temple, Ujjain, across eras

Principal Investigator with three Co-Investigators. Indian Knowledge Systems (IKS) Division, Ministry of Education, Government of India. · ₹8.68 lakh

Completed2022–2023

Review of preliminary design, Statue of Oneness Project, Omkareshwar

Co-Principal Investigator in the Architecture and Planning team. Madhya Pradesh State Tourism Development Corporation (MPSTDC). · ₹25 lakh consultancy

Completed2016

Climate change impact assessment for the urban sector in Madhya Pradesh

Project member. Development Alternatives, New Delhi. Focus areas: Bhopal and Ujjain. · ₹8 lakh

Completed2015

Data collection for Low Carbon Society scenarios for Indian cities

Project member. Government of Japan. Research on fuel consumption, emissions, and technology diffusion. · ₹5 lakh

Patent filed2023

Apparatus for measuring ethanol in motor spirit

Filed with the Controller General of Patents, Designs & Trade Marks, Government of India. Registration number 202321048258.

Completed2025

Smart Buildings and Cities with Remote Sensing & GIS

Authored book published by Taylor & Francis.

Education & professional development

2013

Doctor of Philosophy, Urban Planning

Indian Institute of Technology Roorkee. Thesis: *Strategy for Sustainable Redevelopment of City Core: A Case Study of Ajmer City.*

2009

Master of Urban and Rural Planning (MURP)

Indian Institute of Technology Roorkee. Thesis: *Planning for Public Transportation System for Ajmer City.* CGPA: 8.02/10.

2007

Bachelor of Architecture (B.Arch.)

National Institute of Technology Hamirpur. Thesis: *SOS Children's Village at Dwarka.* CGPA: 7.72/10.

2002

Senior Secondary (Class XII)

Dayanand Bal Niketan Senior Secondary School, Ajmer · Board of Secondary Education Rajasthan · 73.69%.

2000

Secondary (Class X)

Dayanand Bal Niketan Senior Secondary School, Ajmer · Board of Secondary Education Rajasthan · 75.50%.

DIPLOMAS & CERTIFICATES

- **2018:** Post Graduate Diploma in Intellectual Property Rights (PGDIPR), Indira Gandhi National Open University.
- **2017:** Remote Sensing and GIS Applications in Carbon Forestry, Indian Institute of Remote Sensing (IIRS), ISRO.
- **2017:** Climate Change Appreciation Course, Madhya Pradesh Clean Development Mechanism Agency and State Knowledge Management Centre on Climate Change (EPCO).
- **2016–2017:** Sustainable Cities, SDG Academy / Sustainable Development Solutions Network; Certificate of Proficiency with Distinction.
- **2016:** Accredited Professional, Indian Green Building Council (IGBC).

Appointments, teaching & academic service

2023—

Associate Professor, Architecture & Planning

Maulana Azad National Institute of Technology (MANIT), Bhopal.

2010—2023

Assistant Professor, Architecture & Planning

MANIT Bhopal. Joined the institute on 14 June 2010.

2019—2025

Founder Honorary Director, COA-TRC Bhopal Centre

Council of Architecture Training and Research Centre.

ACADEMIC GOVERNANCE & INSTITUTIONAL RESPONSIBILITIES

- Member, Board of Education, Indian Institute of Architects (IIA), Mumbai (2015–2018).
- Member, Board of Studies, AP Goyal University, Shimla (2015 and 2016).
- Executive member, Institute of Indian Interior Designers (IIID), Madhya Pradesh Regional Chapter (2014–2016).
- Executive member, Institute of Urban Transport (IUT), Bhopal Chapter (2014–2016).
- Member, Expert Panel, Rajasthan Public Service Commission (RPSC), Ajmer.
- Member, Standing Committee, All India Council for Technical Education (AICTE).
- MANIT responsibilities: undergraduate admissions (2014–2015), Assistant Proctor (2015–2017), and Undergraduate Ordinance Committee (2016).
- Guided 9 completed postgraduate projects; 1 PhD scholar awarded and 4 PhD scholars in progress.

INVITED LECTURES & EDITORIAL WORK

- **2018 · Keynote speaker:** "Vernacular Architecture and Traditional Construction Practices in Hot and Dry Climate Region," teachers' training programme, MNIT Jaipur with the Council of Architecture.
- **2018 · Resource person:** "Recent Trends & Developments in Civil Engineering," Government College Jhalawar.
- **2020 · Resource person:** COA-TRC Bhopal national online workshop, "The Architectural Thesis: Requirement and Expectations."
- **2017:** Chaired two technical sessions at the national conference "Urban Environmental Management in India: Problems and Prospects," MNIT Jaipur.
- **2013:** Guest Editor, *Journal of the Indian Institute of Architects*, Vol. 78, Issue 12, themed "Traditional Architecture Blending in Modern Design" (ISSN 0019-4913).

Research, consultancy & intellectual property

Ongoing

Study of Design Principles and Indigenous Construction Techniques Used in Mahakaal Temple, Ujjain Across Era

Principal Investigator with three Co-Investigators · ₹8,68,000 · Indian Knowledge Systems (IKS) Division, Ministry of Education, Government of India · 2023–2025.

2022—2023

Review of preliminary design, Statue of Oneness Project, Omkareshwar

Co-Principal Investigator in the Architecture and Planning team · ₹25 lakh consultancy · Madhya Pradesh State Tourism Development Corporation (MPSTDC), Government of Madhya Pradesh.

2016

Climate Change Impact Assessment for the urban sector in Madhya Pradesh

Project member · ₹8 lakh · Development Alternatives, New Delhi · Focus areas: Bhopal and Ujjain.

2015

Data collection for Low Carbon Society scenarios for Indian cities

Project member · ₹5 lakh · Government of Japan · Research on fuel consumption, emissions, and technology diffusion.

2023

Apparatus for measuring Ethanol in Motor Spirit

Patent filing with the Controller General of Patents, Designs & Trade Marks, Government of India · registration number 202321048258 · 19 July 2023.

Books, chapters & study manuals

BOOKS

- Munoth, N., Jain, R. & Dhote, K. K. *Green Cities: Design, Planning and Management.* McMillan Advanced Research Series, McMillan Press. ISBN 9780230332416.
- Munoth, N. & Barbhuiya, M. R. (2020). *Thesis Manual for Bachelor of Architecture.* ISBN 978-93-5426-491-7.
- *Smart Buildings and Cities with Remote Sensing & GIS.* Taylor & Francis, completed in 2025.

BOOK CHAPTERS

1. Barbhuiya, M. R., Munoth, N. & Rajput, R. S. (2020). "Performance of Smart Cities Concerning the Use of Internet of Things: A Case Study of Four Indian Himalayan Cities." In *Smart Cities: A Data Analytics Perspective*, Springer, pp. 257–280. DOI ↗
2. Munoth, N., Tale, M. & Jain, A. (2020). "Reconsidering the Design Features of the Built Environment through Biophilic Principles." The Institution of Engineers (India), Technical Volume IV, pp. 69–75.
3. Munoth, N. & Sharma, N. (2021). "GIS-Based Air Quality Index Spatial Model for Indian Cities." In *Challenges and Solutions for Sustainable Smart City Development*, Springer, pp. 125–167. DOI ↗
4. Munoth, N., Jain, D., Deshmukh, A. & Mandal, N. (2022). "Rethinking Disaster-Resilient Cities: A Close Integration of Planning with Geospatial Technology." In *Advances in Urbanism, Smart Cities, and Sustainability*, CRC Press, pp. 431–448. DOI ↗
5. Munoth, N., Zumu, T. & Gajbhiye, S. (2022). "City of the Indigenous People: A Study of Kohima City, Nagaland, India." In *Indigenous People and Nature*, Elsevier, pp. 61–81. DOI ↗
6. Munoth, N. & Prabha, C. (2022). "Root Causes of Groundwater Contamination for Analyzing Resource Policy Framework and Management." In *Contaminants of Emerging Concerns and Reigning Removal Technologies*, Routledge. DOI ↗
7. Bhattacharya, A., Munoth, N. & Nagaich, A. A. (2022). "The Consequences of Population Growth on Climate Change: A Case of Urbanization Effects vis-à-vis Air Pollutants in Bengaluru, India." In *Social Morphology, Human Welfare, and Sustainability*, Springer. DOI ↗
8. Rajendra, P., Munoth, N. & Sharma, N. (2022). "Urban Intelligence and IoT-UAV Applications in Smart Cities." In *Unmanned Aerial Vehicles and Multidisciplinary Applications Using AI Techniques*, IGI Global, pp. 113–145. DOI ↗
9. Chaurasia, S. & Munoth, N. (2023). "Disaster Risk Reduction: Assessment of Gaps in Policy Framework and Future Scopes in Implementation of Indigenous Practices in India." In *Indigenous Knowledge and Disaster Risk Reduction*, Springer, pp. 465–489. DOI ↗

STUDY MANUALS & TEACHING RESOURCES

- Thesis Manual for Bachelor of Architecture students (copyright 2020).
- Thesis Manual for Bachelor of Planning students (copyright 2019).
- Design Portfolio for Bachelor of Architecture students, Indian Institute of Architects (2015).
- Architectural Appraisal for Bachelor of Architecture students, Indian Institute of Architects (2015).
- Architectural Design Thesis for Bachelor of Architecture students, Indian Institute of Architects (2014).

Journal articles

INTERNATIONAL JOURNALS

1. Bhattacharya, A., Nagaich, A. A. & Munoth, N. (2026). "Studying the '15-minute neighbourhood' approach for inclusive urban landscapes in India: A case of Jaipur, Rajasthan." *Architecture, Structures and Construction*. Article ↗
2. Nagaich, A. A. & Munoth, N. (2025). "Assessment and Strategic Planning for Conservation and Beautification of Shahpura Lake, Bhopal." *Journal of Civil and Environmental Engineering*, 13(11), 60–93.
3. Nagaich, A. A. & Munoth, N. (2025). "Psychological Dimensions of Health and Recreation Concerns in Urban Lake Ecosystems." *TPM – Testing, Psychometrics, Methodology in Applied Psychology*, 32(S6), 1780–1789. Article ↗

4. Nagaich, A. A., Munoth, N. & Pathak, P. (2025). "Restoration Strategies of Urban Lakes for Preventing Public Health Related Concerns." *Journal of Rare Cardiovascular Diseases*, 5(S3), 619–631. Article ↗
5. Nagaich, A. A. & Munoth, N. (2024). "Monitoring and Assessing Wetlands Using Google Earth Engine: A Bibliometric Review (2017–2024)." *International Journal of Environmental Sciences*, 10(6s), 509–525. Article ↗
6. Nair, P., Hajela, A. & Munoth, N. (2022). "The Role of Built Heritage in Sustainable Rural Development: A Key Factor in Spatial Planning of Rural Areas." *Journal of Community Positive Practices*, (3), 126–136. DOI ↗
7. Munoth, N., Nagaich, A. A. & Gehlot, S. (2022). "Transitioning from Wired City to Super City: A Review of Selected 'Smart City' Case Studies." *GeoJournal*. DOI ↗
8. Nair, P., Singh, D. P. & Munoth, N. (2022). "'Heritage Village' Designation Impact on the Village: A Case Study on the Perception of the Residents of Kalpathy Village (Kerala)." *Journal of The Institution of Engineers (India): Series A*, 103(2), 493–500. DOI ↗
9. Nair, P., Singh, D. P. & Munoth, N. (2022). "The Role of Built Heritage in Sustainable Rural Development: A Key Factor in Spatial Planning of Rural Areas." *International Transaction Journal of Engineering, Management, & Applied Sciences & Technologies*, 13(4), 13A4C, 1–10. DOI ↗
10. Chaurasia, S. & Munoth, N. (2022). "Carbon Sequestration Potential of Urban Trees: A Case of Kolar Area in Bhopal City, India." *Journal of The Institution of Engineers (India): Series A*, 103(2), 359–374. DOI ↗
11. Nair, P., Singh, D. P. & Munoth, N. (2021). "Challenges and New Paradigms in Conservation of Heritage-Based Villages in Rural India: A Case of Pragpur and Garli Villages in Himachal Pradesh." *Advances in Science, Technology and Engineering Systems Journal*, 6(1), 1112–1119. DOI ↗
12. Munoth, N. & Nagaich, A. A. (2015). "Planning Interventions for Lake Conservation: A Case of Shahpura Lake, Bhopal." *Journal of The Institution of Engineers (India): Series A*, 96(3), 193–205. DOI ↗
13. Munoth, N., Kamal, M. A. & Brar, T. S. (2015). "Building Design Techniques to Reduce Impact of Global Warming." *Journal of Environmental Science, Technologies and Engineering*, XV, 15–20.
14. Munoth, N. & Brar, T. S. (2014). "Solar and Green Building Guidelines for Hot Arid Climate in India." *Building Research Journal*, 61(1), 59–65.
15. Munoth, N., Jain, R. K., Raheja, G. & Brar, T. (2013). "Issues of Sustainable Redevelopment of City Core: Study of Developed and Developing Countries." *Journal of The Institution of Engineers (India): Series A*, 92(2), 117–122. DOI ↗

NATIONAL JOURNALS

- Daniel, M. J. & Munoth, N. (2015). "Professional Ethics in Town Planning: A Case of Developed and Developing Countries." *Institute of Town Planners India*, pp. 33–49.
- Munoth, N. & Nagaich, A. (2013). "Reflections on Education and its Purpose: Implications for Teacher Education." *Indian Journal of Higher Education*, 4(2), 110–117.
- Munoth, N. & Kumar, A. (2011). "Vernacular Architecture: A Prerequisite for Sustainable Development." *Architecture Time Space & People*, Council of Architecture, pp. 16–22.

Conference papers & proceedings

INTERNATIONAL CONFERENCE PROCEEDINGS

- Munoth, N. & Nagaich, A. (2018). "Planning for Reducing Environmental Impacts in Peri-Urban Areas: A Case of Gaya, India." 2nd Asian Regional Conference on Peri-Urbanization, Pusan National University, Busan, South Korea.

- Munoth, N. & Bimrah, K. (2018). "Physical Development Strategies for a Ward through Indicator-Based Analysis: A Case of Bhopal City, Madhya Pradesh, India." 4th Annual International Conference on Urban Planning and Property Development, Singapore, pp. 1–9.
- Munoth, N. & Rathore, S. (2018). "Ward-wise Planning and Distribution of Parks and Gardens Using Network and Spatial Analysis: Case Study of Indore, India." International Conference for Sustainable Design of the Built Environment (SDBE 2018), pp. 659–664.
- Munoth, N., Jain, D. & Anwer, M. (2018). "The Integrated Approach Towards Inner-City Sustainability in Today's Context: Nagpur, India." International Conference on Urban Sustainability (ICUS), Jaipur. Paper ↗
- Munoth, N. & Thakur, S. (2017). "An Approach to Make Indian Cities Sustainable through the Concept of Green-Blue Infrastructure." *Urbanistica Informazioni* special issue, pp. 989–1000.
- Munoth, N. (2016). "Understanding Existing and Prospective Issues of Ajmer City Core, Rajasthan, India." BRICS International Forum, New Delhi.
- Munoth, N. & Barbhuiya, M. R. (2015). "Redevelopment Strategies of Ajmer City Core for Safe, Sustainable and Smart Heritage City." Municipalika, Jaipur.
- Tiwari, H., Dhote, K. K. & Munoth, N. (2015). "Sustainable Urban Growth of Old City Area: Bhopal, India." Green and Blue Infrastructures, Naples, pp. 201–204.
- Kumar, A., Vihari, V. R. S. & Munoth, N. (2015). "New Approach of Advancing Green-Blue Infrastructure: Bhopal, India." Green and Blue Infrastructures, Naples, pp. 62–67.
- Barbhuiya, M. R., Garg, Y. K. & Munoth, N. (2015). "Developing Guidelines for a Smart City in India Using Garden City Theory as a Tool." Green and Blue Infrastructures, Naples, pp. 12–14.
- Munoth, N., Pankaj, S. & Jindal, D. (2015). "Re-Calibration of the Professed 'Smart' Neighbourhood." Green and Blue Infrastructures, Naples, pp. 49–51.
- Munoth, N. & Garg, Y. K. (2013). "Dynamics of Housing Improvement Process of the Urban Poor – An Approach." Asia Pacific Network for Housing Research, University of Malaya, Kuala Lumpur.
- Munoth, N. (2010). "Urban Traffic Characteristics in India: A Case Study of Ajmer City." International Conference on Design & Construction Transport Structures, Hyderabad, pp. 293–310.
- Munoth, N., Jain, R. K. & Raheja, G. (2010). "Pedestrian Movement in City Core Area: A Case Study of Roorkee, India." TRANSTEC, New Delhi, pp. 210–219.

NATIONAL CONFERENCE & SYMPOSIUM PROCEEDINGS

- Vihari, V. R. S., Sunkara, P. & Munoth, N. (2014). "Sustainability and Cosmic Energy of Indian Temples." National Symposium on Sustainability and the Built Environment, Bengal Engineering and Science University, pp. 130–135.
- Sharma, G. P., Shukla, P. & Munoth, N. (2013). "Reflections on Education and its Purpose." National Seminar on Education for Change, Navrachana University, Vadodara.
- Sharma, G. P., Siddiqui, A. H. & Munoth, N. (2013). "Re-conceptualizing the High Rise: Scenario of Bhopal." National Conference on Recent Advances in Civil Engineering, NIT Patna, pp. 237–247.
- Singh, K., Soni, J. & Munoth, N. (2012). "Wind Simulation and Solar Simulation of Building Using Computer Aided Software Autodesk Vasari: An Overview." IBPSA National Convention and Conference, MNIT Jaipur.
- Sharma, G. P., Aggarwal, S. & Munoth, N. (2012). "Sustainability Concept in Arid Region: A Case Study of Mandawa, Rajasthan." National Conference on Sustainable Cities in Arid Region, MNIT Jaipur.
- Munoth, N. & Jain, R. K. (2010). "Transportation Responsive Land Use Planning—Ajmer, India." National Seminar on Sustainable Lifelines, ITPI, pp. 105–113.

- Munoth, N., Jain, R. K. & Munoth, P. (2010). “Urban Water Resource Management in India.” National Seminar on Sustainable Management and Conservation of Water Resources, Rajasthan Technical University, pp. 57–64.
- Jain, R. K. & Munoth, N. (2009). “Urban Transportation in India—Issues and Strategies.” National Conference on Innovation Without Limits, NIT Hamirpur, pp. 283–289.

Conference contributions listed here: 14 international and 8 national conference and symposium papers.

Awards, memberships & skills

- **2023 · K. F. Antia Memorial Prize,** The Institution of Engineers (India), for the paper on carbon sequestration potential of urban trees in Kolar, Bhopal.
- **2022 · 13th CIDC Vishwakarma Award, Academician category,** Construction Industry Development Council, New Delhi.
- **2019 · Young Engineer Award, Architectural Engineering,** The Institution of Engineers (India), 35th National Convention of Architectural Engineers, Thrissur.
- **2016 · Prof. R. C. Singh Prize,** The Institution of Engineers (India), for the Shahpura Lake conservation paper.
- **2006 · GATE,** All India Rank 83 in Architecture and Planning.
- **NCC:** B and C certificates.

PROFESSIONAL MEMBERSHIPS

- Council of Architecture (COA) · Life Member, 2007.
- Institute of Town Planners, India (ITPI) · Associate Life Member, 2011.
- Indian Institute of Architects (IIA) · Associate Life Member, 2012.
- Institute of Indian Interior Designers (IIID) · Associate Life Member, 2012.
- Institute of Urban Transport (IUT) · Life Member, 2013.
- Indian Building Congress (IBC) · Life Member, 2013.
- Indian Institute of Valuers (IIV) · Life Member, 2015.
- Indian Green Building Council (IGBC) · Life Member (Accredited Professional), 2015.
- Indian Society of Remote Sensing (ISRS) · Life Member, 2020.
- Indian Society of Geomatics (ISG) · Life Member, 2020.
- International Society of City and Regional Planners (ISOCARP) · Life Member, 2020.
- The Institution of Engineers (India) (IEI) · Life Member, 2021.
- Institute of Electrical and Electronics Engineers (IEEE) · Member, 2021.

PROFESSIONAL & TECHNICAL SKILLS

Team management · Design development · Research · Project feasibility · Project management and planning · Construction · MS Office · Adobe Photoshop · Autodesk AutoCAD · Autodesk 3ds Max · Autodesk Revit · Google SketchUp

Web of Science Researcher ID: A-1234-2016 · Scopus Author ID: 57195776608 · Languages: English, Hindi, Sanskrit